Alexander M. **Long**

Scientist 3 · Material Science and Technology (MST) · Los Alamos National Laboratory

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Publication h-index: 15 · Citizenship: United States of America

Education

University of Notre Dame

Notre Dame, Indiana USA

Ph.D., Physics — *(June 2009 - July 2016)*

- Thesis Topics: An Indirect Study of The Astrophysical ^{34}Ar(α,p)^{37}K Reaction and its Influence on Type-1 X-Ray Burst Light Curves.
- Advisor: Professor Michael Wiescher

Florida State University

Tallahassee, Florida USA

B.S., Physics with Honors — *(August 2004 - May 2009)*

- Honors Thesis Topic: Time-of-Flight Calibrations of Neutron Wall Array at John D. Fox Superconducting Accelerator Laboratory
- Advisor: Professor Grigory Rogachev (Now at Texas A&M University)

Research Experience

Lujan Neutron Scattering Center and Weapons Neutron Research Facility

LANSCE @ LANL, USA

Instrument Scientist for FP5 and FP11 — *Dec. 2018 - Present*

- **Primary liaison for neutron imaging projects at LANSCE**
  - Leading & supporting all neutron imaging measurements on FP4, FP5, FP11, and FP60R beamlines.
  - Enabling world-class advanced neutron imaging capabilities through developing and commissioning several novel neutron imaging detectors, sample environments, and data analysis codes.
  - Fostering growth of neutron imaging as LANSCE through collaboration meetings, invited talks and seminars.
  - Actively developing new neutron imaging techniques to meet LANL's scientific missions.
  - PI or Co-PI on several LANL-LDRD or DOE-NE funded projects (See "Awarded Funding" section).
  - Converted all motion control to Zaber motion stages and developed python based motion control programs that are integrated with the LANSCE EPICS framework.
- **Developed Energy Resolved Neutron Imaging (ERNI) capabilities on FP5/ERNI**
  - Developed ERNI techniques such as neutron resonance imaging (NRI) and Bragg-edge imaging (BEI) on FP5.
  - Mentoring grad students in developing a new state-of-the-art neutron resonance imaging analysis code **TRINIDI**.
  - Led the execution of first-of-a-kind NRI measurements on irradiated nuclear fuel pellet.
  - Actively participating in rebuild of FP12 for future ERNI-based post-irradiation examination capability.
  - Developing open-sourced user-friendly python package **PLEIADES** for neutron resonance imaging analysis with SAMMY.
- **Developed ERNI capabilities with photon-counting neutron imaging technology at LANSCE**
  - Led effort to bring photon-counting TPX3Cams to Lujan Center and WNR. Now have three TPX3Cam based systems that can operate on four different flight paths.
  - Lead scientist in developing TPX3Cam data acquisition and analysis code **HERMES**.
  - Developing machine learning algorithms to enable more advanced and robust neutron imaging analysis of TPX3 data.
  - Collaborating with scientist in Physics- and NEN-divisions to introduce Timepix technology to respective imaging efforts.
- **Developed novel neutron imaging technique to measure thermophysical properties of molten salts and uranium metals**
  - Developed density measurement technique called Density via Neutron Radiography (DvNR)
  - Successfully applied and published use of DvNR technique to measure densities of Uranium and Plutonium based salts
  - Part of team to bring first-of-a-kind DOE-NE GAIN funding to LANL in collaborations with TerraPower
  - Lead scientist in applying DvNR to measure densities of molten uranium and uranium-niobium metals
- **Developed hydrogen mapping capabilities using neutron imaging techniques**
  - Developed neutron imaging capabilities at Lujan Center to probe temperature driven hydrogen diffusion in metal hydrides.
  - Successfully published reports and publications for DOE-NE Microreactor program.
  - Led design and commissioning of state-of-the-art compact dual zone furnace.
  - Built pipelines for neutron imaging based hydrogen mapping capabilities with Colorado School of Mines.
- **Developed fast (MeV) neutron imaging and scintillation characterization capabilities on Flight Path 60R at WNR**
  - Co-led NNSA DNN NA-22 programmatic work on 60R to characterize nano-guide scintillator materials for fast neutron imaging.
  - Effectively showcased event-mode imaging as a Fast Neutron Resonance Radiography technique using MeV neutrons.
  - Successfully demonstrated the use of TPX3Cams to observe mm sized proton recoil tracks in fast plastics.
- **Developed Bragg edge and grain-mapping capabilities on FP4/HIPPO and FP5/ERNI at Lujan Center**
  - Co-lead design and commission of first-ever simultaneous diffraction and transmission measurements at LANSCE on FP4.
  - Performed simultaneous NRI and diffraction measurements on irradiated UO_2 sample on FP4/HIPPO.
  - Performed several Bragg edge CT scan on various AM sample on FP5/ERNI using TPX3Cams.
- **Fostered and Developed Neutron Grating Interferometry (nGI) capabilities on FP11/ASTERIX.**
  - Built collaborations of nGI expertise with scientist from NIST and PSI.
  - Supported several external nGI measurements to investigate pore sizes in AM metals and various concrete mixtures.
  - Part of collaboration to successfully propose a new neutron imaging beamline, CUPI^{2}D, for the Second Target Station (STS).

Weapons Neutron Research Facility

LANSCE @ LANL, USA

Post Doctoral Researcher

Sept. 2016 - Dec. 2018

- **Assisted with development of the Low Energy (N,Z) (LENZ) experimental program at WNR/LANSCE.**
  - Developed digital data acquisition systems, along with unpacking and analysis codes for the LENZ experimental program.
  - Developed and operated several (n,Z) experimental setups on FP15R for the LENZ program.
  - Presented nuclear data cross-section measurements at Cross Section Evaluation Working Group (CSEWG)
- **Investigated H production reaction ^{55}Mn(n,p)^{55}Cr for core structural materials to be used in future nuclear reactor designs.**
  - Performed cross-section measurements on the ^{55}Mn(n,p)^{55}Cr reaction using LENZ to investigate the influence of this reaction as a source of neutron irradiation damage in structural materials in future fission and fusion devices.
- **Investigations of neutron irradiation damage in F-M steels through precision measurements of the He gas production reaction ^{56}Fe(n,α).**
  - Performed ^{56}Fe(n,α) and ^{52}Cr(n,α) reaction cross-section measurements using LENZ to better understand He production rates in various F-M steels material candidates considered for use in future reactor core designs.

Mentoring

2025-Pres.	**Nicholas Mendez (Post-Doc)**, LANL	*Mentor*
2024-Pres.	**Adam Farren (Undergraduate)**, Colorado School of Mines	*Mentor*
2023-2024	**Tim Jaeger (Graduate Student)**, Technical University of Darmstadt	*Co-Mentor*
2022-2023	**James Torres (Post-Doc)**, Now Insturment Scientist on MARS beamline at HFIR/ORNL	*Mentor*
2020-2023	**Thilo Balke (Graduate Student)**, Now Research Scientist at Samsung	*Co-Mentor*
2020-2023	**Danielle Schaper (Graduate Student)**, Now Post-doc in P-1	*Co-Mentor*
2021-2023	**Daniel Eigenbach (Post-bach Student)**, Now Research Technologist in MST-8	*Co-Mentor*
2019-2020	**Darcy Newmark (Undergraduate Student)**, Now Graduate Student at MIT	*Mentor*

Awarded Funding

2025-2027	**Development of Highly Portable Mixed Radiography Systems with Timepix**, PI	*DOE LDRD*
2025	**Demonstrating Timepix Sensors for Enhanced Neutron Detection and Non-Destructive Monitoring**, PI	*LANL PPO-EMI*
2024-2025	**Advanced Characterization of Metal Hydride Moderators with Pulsed Neutrons**, Co-PI	*LANL LDRD-MFR*
2024	**Nuetrons for Nuclear: NSUF @ LANSCE**, PI	*INL NSUF*
2024	**Implementation and Demonstration of 2D and 3D crystal and microstructure measurements using pulsed neutrons**, Co-PI	*LANL PPO-EMI*
2023-2024	**Thermophysical Density Measurements of Molten Uranium Niobium Alloys**, PI	*LANL LDRD-DI*
2023-2024	**Strengthening LANL's Radiography Applications via Event-Mode Imaging and ML Techniques**, Co-PI	*LANL LDRD-ER*
2022-2025	**Development of Hydrogen Transport Models for High Temperature Metal Hydride Moderators**, Co-PI	*DOE-NE NEUP*
2021-2023	**Density Measurements of Plutonium Bearing Salts via Neutron Beam Dilatometry**, Co-I	*DOE-NE GAIN*
2020-2023	**In-situ Spatial Mapping of Hydrogen in Yttrium Hydrides at LANSCE**, Co-PI	*DOE-NE MRP*
2021-2023	**Advanced Characterization to Enable Prediction of Actinide-Molten Salt Behavior**, Co-I	*LANL LDRD-DR*
2021	**Exploring Safeguard Signatures with Energy Resolved Neutron Imaging for Future Molten Salt Reactor Designs**, PI	*LANL LDRD-MFR*
2019-2022	**Prioritizing the Prior: Advanced Inversion Algorithims for Scientific Data Analysis**, Co-I	*LANL LDRD-DR*

Organizations, Committees, and Positions

2024-Pres.	**Vice-Chair**, Local Sections Committee	*ANS*
2024-Pres.	**Reviewer**, Nature Scientific Reports	*ASC*
2023-Pres.	**LANL Point of Contact**, Nuclear Science User Facilities	*INL*
2023	**Organizing Member**, Computational Imaging XXI Conference	*IS&T*
2022-Pres.	**Reviewer**, Journal of Industrial & Engineering Chemistry Research	*ASC*
2021-Pres.	**Reviewer**, Journal of Imaging; Crystals; and Quantum Beam Science	*MDPI*
2016	**Vice Chair**, Frontiers in Nuclear Astrophysics Meeting Organizing Committee	*JINA-CEE*
2010-2011	**Committee Member**, Department of Physics Graduate Recruitment Committee	*UND*
2011-2014	**Board Member**, Graduate Physics Students Conference Committee	*UND*

Honors

2023	**Distinguished Performance Award: Hydride Moderator Development Team**,	*LANL*
2016	**Recipient of the Nuclear Science Laboratory's Cornelius P. Browne Memorial Award**,	*UND*
2014	**Notre Dame Graduate Student Union Conference Presentation Award**,	*UND*
2014	**Notre Dame Graduate School Professional Development Award**,	*UND*
2008	**Guenter Schwarz Memorial Scholar Award**,	*FSU*

Technical Experience

Imaging Systems	CCD, CMOS, and sCMOS cameras
	Timepix2 & Timepix3 based imaging cameras
	Optical systems: scintillators, lenses, and image intensifiers
Radiation Detection	Fast Plastics + PMT Neutron Detectors
	Silicon Detectors: Diodes and Double sided
	High Purity Germanium Detectors
	^{3}He Proportional Counters
	Multi-Wire Drift Chambers
	Mircochannel Plate Detectors
Analog Pulse Processing	Pre-Amplifiers, Constant Fraction Discriminators, Amplifiers, Gate-Generators
Digital Pulse Processing	Timepix3 SPIDR Readout boards
	CAEN Family Digitizers
Ion Beam Production	Multi-Cathode Source of Negative Ions by Cesium Sputtering @ NSL: Operations
	Helium Ion Source @ NSL: Operations
	Electron Cyclotron Resonance Ion Source @ NSL: Operations and maintenance
Ion Beam Transportation	10 MV FN Tandem Accelerator @ NSL: Operations and maintenance
	5 MV Van der Graaf Accelerator @ NSL: Operations and maintenance
	Beamline optics and fabrication
	Dispersion matching of beam lines to magnetic spectrographs
Vacuum Systems	Roughing pumps, Roots Blowers, Turbo-molecular Pumps, Cryogenic pumps

Computational Experience

Base Languages	Python, C/C++, BASH & ZSH, HTML/CSS, Java, LaTeX
Proficient Python Libraries	NumPy, SciPy, Pandas, Matplotlib, Pillow, Scikit Image, Scikit Learn, Tomopy, pyEpics, pyTorch
Macro Programing	ImageJ (Java), ROOT (C/C++), Qt (python), CUDA(C)
Modeling & Simulations	Geant4, SAMMY, SRIM, TALYS, XNet, DWUCK4, AutoCAD

Seminars and Talks (invited)

Frontiers in Neutron Radiography — *Seattle, WA*
Optica Imaging Congress: Radiographic Imaging and Tomography (RadIT) — *2025*

Harnessing Event-Based Imaging Systems for Fast Neutron Radiography at LANSCE — *Fort Worth, TX*
27th International Conference on the Application of Accelerators in Research and Industry (CAARI) — *2024*

Neutrons for Nuclear: NSUF @ LANSCE — *Gainville, FL*
Department of Materials Science and Engineering, University of Florida — *2024*

Sifting Through Light: A Need for Efficient Processing and Reconstruction in Photon-Counting Based Neutron Imaging Data — *Cleveland, OH*
Materials Data Science for Stockpile Stewardship Center of Excellence, Case Western University — *2024*

Neutron PIE: Accelerating Post-Irradiation Examination with Advanced Neutron Imaging for Next-Gen. Nuclear Reactor Materials — *South Bend, IN*

Nuclear Science Seminar, Department of Physics, University of Notre Dame — *2023*

Neutron Imaging at LANSCE: Characterizing Future Materials for Next Gen. Reactors — *Pittsburgh, PA*

Materials Science & Technology — *2022*

Characterizing Materials for Next Generation Nuclear Reactors using Neutron Imaging — *Virtual*

University of Las Vegas Nevada, NSST Consortium Seminar — *2022*

Neutron Imaging at LANSCE — *Virtual*

TMS Annual Meeting and Exhibition — *2021*

Publications

Public Communications

'Neutron Vison as Los Alamos: Exploring the Frontiers of Nuclear Materials Science' **Alexander Long** & Sven Vogel, *Nuclear News 68-2 (2025)*

'Crystal Clear: High-fidelity neutron radiography is the future of neutron science and materials characterization' **Alexander Long** & Sven Vogel, *1663 Magazine, (**submitted**)*

Peer-Reviewed

'Real-Time Neutron Imaging for Hydrogen Migration in Subsurface Lithologies' Prakash Purswani, Bijay K C, James Torres, Yuxuan Zhang, **Alexander Long**, Chelsea W. Neil, Eric Guiltinan, Hakim Boukhalfa, Tim Germann, & Michael Gross, *Environmental Science & Technology Letters, (**submitted**)*

'Impact of extinction effects on neutron transmission and diffraction in solid beryllium metal' S. Xu, D.D. DiJulio, J.I. Marquez Damian, S.C. Vogel, **A.M. Long**, T.Y. Hirsh, T. Kittelmann, V. Kuksenko, & G. Muhrer, *Journal of Applied Crystallography, (**accepted**)*

'Energy-Resolved Neutron Imaging and Diffraction including Grain Orientation Mapping Using Event Camera Technology' Tsviki Y. Hirsh, Andrew F.T. Leong, **Alexander M. Long**, Adrian S. Losko, Alexander Wolfertz, Daniel J. Savage, Tim T. Jaeger, John Rakovan, James J. Wall, & Sven C. Vogel, *Nature Scientific Reports*, 15 12901 (2025)

'Energy-Resolved MeV Neutron Analysis of a Mock-up Nuclear Waste Package' Tim T. Jaeger, Tsviki Y. Hirsh, Stefan Scheuren, **Alexander M. Long**, Adrian S. Losko, Alexander Wolfertz, Marc Zimmer, Markus Roth, & Sven C. Vogel, *Nature Scientific Reports*, 15 823 (2025).

'Exploring interface dynamics in plastic bonded-explosives: synthesis, characterization, and neutron reflectometry' Michael R. Bull, Matthew J. Herman, Erik B. Watkins, John D. Yeager,Michael P. Marquez, Alexander M. Long, D. Travis Carver,Robert Knepper, & Tonya L. Kuhl, *Journal of Energetic Materials*, 1-24 (2025).

'Large Field-of-View Event-Mode Camera for High-Precision Epithermal Neutron Resonance Imaging' Tsviki Y. Hirsh, **Alexander M. Long**, Adrian S. Losko, Tim T. Jaeger, Alexander Wolfertz, & Sven C. Vogel, *Nature Scientific Reports*, 15 14026 (2025).

'Advances in Detection for Neutron Reflectometry with Time-Resolved Imaging Detectors' Anton Khaplanov, John Ankner, Mathtew Loyd, Erik Watkins, Tsviki Hirsh, Adrian Losko, Alexander Wolfertz, **Alexander Long**, Sven Vogel, and G. Jeffrey Sykora, *Nature Scientific Reports*, 15 25014 (2025).

'Energy-Resolved Fast-Neutron Radiography Using an Event-Mode Neutron Imaging Detector' Alexander Wolfertz, Adrian Losko, **Alexander M. Long**, Sophia Brodish, Aaron E. Craft, Anton Khaplanov, Sven C. Vogel, Ronald O. Nelson, Stephen A. Wender, Anton Tremsin, Tsviki Y. Hirsh, Tim T. Jager, Manuel Morgano, & Patrick Feng, *Nature Scientific Reports*, 14 30487 (2024).

'LumaCam: A Novel Class of Position-Sensitive Event Mode Particle Detectors using Scintillator Screens' Alexander Wolfertz, Alex Gustschin, Michael Schulz, **Alexander M. Long**, Anton Khaplanov, Tsviki Y. Hirsh, Andrei Nomerotski, Manuel Morgano, Anton Tremsin, Giacomo Mauri, G. Jeff Sykora, & Adrian Losko, *Nature Scientific Reports*, 14 30495 (2024).

'Low Energy Neutron-induced Charged-particle (Z)(LENZ) instrument development with a focus on pulse shape discrimination for low-energy charged particles' Hye Young Lee, S.A. Kuvin, S.M. Mosby, C.J. Prokop, **A.M. Long**, D. Votaw, A. Georgiadou, J. Görres, E. Stech, & M. Wiescher, *Nuclear Instruments and Methods in Physics Research* 1069 (2024).

'Small-angle scattering and dark field imaging for validation of a new neutron far-field interferometer' Caitlyn M. Wolf, Peter Bajcsy, Wei-Ren Chen, Robert M. Dalgliesh, M. Cyrus Daugherty, Liliana de Campo, Fumiaki Funama, Lilin He, Michael Huber, David Jacobson, Paul Kienzle, Youngju Kim, Hubert King, Nikolai Klimov, Jacob LaManna, Fankang Li, **Alexander M. Long**, Ryan Murphy, Gergely Nagy, Sarah M. Robinson, Pushkar Sathe, Gregory N. Smith, Anna Sokolova, Sven C. Vogel, Erik B. Watkins, Yuxuan Zhang, Dan Hussey, & Katie Weigandt, *Journal of Applied Crystallography* 57(6), (2024).

'Phase and contrast moire signatures in two-dimensional cone beam interferometry' D. Sarenac, G. Gorbet, Charles W. Clark, D. G. Cory, H. Ekinci,2 M. E.Henderson, M. G. Huber, D. Hussey, C. Kapahi, P. A. Kienzle, Y. Kim, **A. M. Long**, J. D. Parker, T. Shinohara, F. Song, & D. A. Pushin, *Physical Review Research*, 6, L032054, (2024).

'A compact furnace to support in situ neutron imaging of hydrogen dynamics in yttrium hydride moderators' James Torres, **Alexander M. Long**, D. Travis Carver, Christopher Matthews, Sven Vogel, Tyler Smith, Caitlin Kohnert, Erik Luther, Holly Trellue and Aditya Shivprasad, *MRS Advances* (2024).

'Demonstration of neutron time-of-flight diffraction with an event-mode imaging detector' Tim T Jäger, Adrian S Losko, Alexander Wolfertz, Søren Schmidt, Mads Bertelsen, Anton Khaplanov, Sean R Agnew, Fumiaki Funama, Manuel Morgano, Markus Roth, Jason R Gochanour, **Alexander M. Long**, Luca Lutterotti, & Sven C Vogel, *Journal of Applied Crystallography*, 57(4), (2024).

'TRINIDI: Time-of-Flight Resonance Imaging with Neutrons for Isotopic Density Inference' T. Balke, **A.M. Long**, S.C. Vogel, B. Wohlberg, and C.A. Bouman, *IEEE Transactions on Computational Imaging* vol. 10, pp. 154-169, (2024).

'The Complex, Unique and Powerful Imaging Instrument for Dynamics (CUPI2D) at the Spallation neutron Source' A. Brugger, H.Z. Bilheux, J.Y.Y. Lin, G.J. Nelson, A. Kiss, D.J.P. Morris, M. Connolly, **A.M. Long**, A.S. Tremsin, A. Strzelec, M. Anderson, R.J. Agasie, C.E.A. Finney, M.L. Wissink, M.H. Hubler, R. Pellenq, C.E.White, B.J. Heuser, A. Craft, J.M. Harp, C. Tan, K. Morris, B, Schillinger, & S.C.Vogel *Review of Scientific Instruments* 94, 051301 (2023).

'Density Measurement of NaCl-MgCl2-PuCl3 and NaCl-UCl3-PuCl3 Molten Salt Systems by Neutron Radiographic Dilatometry' Stephen Scott Parker, Alexander M. Long, Charles Lhermitte, Sven Vogel, Marisa Monreal and J. Matthew Jackson, Journal of Molecular Liquids, 346 (2022).

'Remote Density Measurements of Molten Salts via Neutron Radiography.' **Long, A. M.**, Parker, S. S., Carver, D. T., Jackson, J. M., Monreal, M. J., Newmark, D. A., & Vogel, S. C. *Journal of Imaging*, 7(5), 88 (2021).

'Effects of Hydrogen Redistribution at High Temperatures in Yttrium Hydride Moderator Material.' Trellue, H. R., **Long, A. M.**, Luther, E. P., Carver, D. T. & Mehta, V. K. *The Journal of The Minerals, Metals & Materials Society*, 73, 3513-3518 (2021).

'Flexible 3D printed silicones for gamma and neutron radiation shielding.' Talley, S. J., Robison, T., **Long, A. M.**, Lee, S. Y., Brounstein, Z., Lee, K. S., & Labouriau, A., *Radiation Physics and Chemistry*, 188, 109616 (2021).

'α-unbound levels in ^{34}Ar from $^{36}Ar(p,t)^{34}Ar$ reaction measurements and implication for the astrophysical $^{30}S(\alpha,p)^{33}Cl$ reaction rate'. **A.M. Long**, T. Adachi, M. Beard, G. P. A. Berg, M. Couder, R. J. deBoer, M. Dozono, J. Görres, H. Fujita, Y. Fujita, K. Hatanaka, D. Ishikawa, T. Kubo, H. Matsubara, Y. Namiki, S. O'Brien, Y. Ohkuma, H. Okamura, H. J. Ong, D. Patel, Y. Sakemi, Y. Shimbara, S. Suzuki, R. Talwar, A. Tamii, A. Volya, T. Wakasa, R. Watanabe, M. Wiescher, R. Yamada, & J. Zenihiro, *Physical Review C* 97, 054613 (2018).

'Determination of $^{20}Ne(p,\gamma)^{21}Na$ cross sections from E_p = 500 - 2000 keV'. S. Lyons, J. Gorres, R.J. deBoer, E. Stech, Y. Chen, G. Gilardy, Q. Liu, **A.M. Long**, M. Moran, D. Robertson, C. Seymour, B. Vande Kolk, & M. Wiescher, *Physics Review C* 97 (2018).

'An indirect study of the stellar $^{34}Ar(\alpha,p)^{37}K$ reaction rate through $^{40}Ca(p,t)^{38}Ca$ reaction measurements' **A.M. Long**, T. Adachi, M. Beard, G. P. A. Berg, Z. Buthelezi, J. Carter, M. Couder, R. J. deBoer, R. W. Fearick, S. V. Förtsch, J. Göres, J. P. Mira, S. H. T. Murray, R. Neveling, P. Papka, F. D. Smit, E. Sideras-Haddad, J. A. Swartz, R. Talwar, I. T. Usman, M. Wiescher, J. J. Van Zyl, and A. Volya *Physical Review C* 95, 055803 (2017).

'Probing astrophysically important states in the ^{26}Mg nucleus to study neutron sources for the s-process'. Talwar, R., Adachi, T., Berg, G.P.A., Bin, L., Bisterzo, S., Couder, M., DeBoer, R.J., Fang, X., Fujita, H., Fujita, Y., Gorres, J., Hatanaka, K., Itoh, T., Kadoya, T., **Long, A.**, Miki, K., Patel, D., Pignatari, M., Shimbara, Y., Tamii, A., Wiescher, M., Yamamoto, T., Yosoi, M. *Physics Review C* 93 (2016).

'Low energy neutron background in deep underground laboratories'. Best, A., Gorres, J., Junker, M., Kratz, K.-L., Laubenstein, M., **Long, A.**, Nisi, S., Smith, K., Wiescher, M. *Nuclear Instruments and Methods in Physics Research* 812 (2016).

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'Systematic study of (α,γ) reactions for stable nickel isotopes'. Simon, A., Beard, M., Spyrou, A., Quinn, S.J., Bucher, B., Couder, M., DeYoung, P.A., Dombos, A.C., Gorres, J., Kontos, A., **Long, A.**, Moran, M.T., Paul, N., Pereira, J., Robertson, D., Smith, K., Stech, E., Talwar, R., Tan, W.P., Wiescher, M. *Physics Review C* 92 (2015).

'First Direct Measurement of C12(C12,n)Mg23 at Stellar Energies'. Bucher, B., Tang, X.D., Fang, X., Heger, A., Almaraz-Calderon, S., Alongi, A., Ayangeakaa, A.D., Beard, M., Best, A., Browne, J., Cahillane, C., Couder, M., Deboer, R.J., Kontos, A., Lamm, L., Li, Y.J., **Long, A.**, Lu, W., Lyons, S., Notani, M., Patel, D., Paul, N., Pignatari, M., Roberts, A., Robertson, D., Smith, K., Stech, E., Talwar, R., Tan, W.P., Wiescher, M., Woosley, S.E. *Physical Review Letters* 114 (2015).

'First application of the γ-summing technique in inverse kinematics'. Quinn, S.J., Spyrou, A., Simon, A., Battaglia, A., Bowers, M., Bucher, B., Casarella, C., Couder, M., Deyoung, P.A., Dombos, A.C., Greene, J.P., Gorres, J., Kontos, A., Li, Q., **Long, A.**, Moran, M., Paul, N., Pereira, J., Robertson, D., Smith, K., Smith, M.K., Stech, E., Talwar, R., Tan, W.P., Wiescher, M. *Nuclear Instruments and Methods in Physics Research* 575 (2014).

'Measurement of the 58Ni(α,γ)62Zn reaction and its astrophysical impact'. Quinn, S.J., Spyrou, A., Bravo, E., Rauscher, T., Simon, A., Battaglia, A., Bowers, M., Bucher, B., Casarella, C., Couder, M., Deyoung, P.A., Dombos, A.C., Gorres, J., Kontos, A., Li, Q., **Long, A.**, Moran, M., Paul, N., Pereira, J., Robertson, D., Smith, K., Smith, M.K., Stech, E., Talwar, R., Tan, W.P., Wiescher, M. *Physics Review C* 89 (2014).

'Measurement of the 90,92Zr(p,γ)91,93Nb reactions for the nucleosynthesis of elements near A=90'. Spyrou, A., Quinn, S.J., Simon, A., Rauscher, T., Battaglia, A., Best, A., Bucher, B., Couder, M., Deyoung, P.A., Dombos, A.C., Fang, X., Gorres, J., Kontos, A., Li, Q., Lin, L.Y., **Long, A.**, Lyons, S., Meyer, B.S., Roberts, A., Robertson, D., Smith, K., Smith, M.K., Stech, E., Stefanek, B., Tan, W.P., Tang, X.D., Wiescher, M. *Physics Review C* 88 (2013).

'Testing the mutually enhanced magicity effect in nuclear incompressibility via the giant monopole resonance in the 204,206,208Pb isotopes'. Patel, D., Garg, U., Fujiwara, M., Adachi, T., Akimune, H., Berg, G.P.A., Harakeh, M.N., Itoh, M., Iwamoto, C., **Long, A.**, Matta, J.T., Murakami, T., Okamoto, A., Sault, K., Talwar, R., Uchida, M., Yosoi, M. *Physics Letters B* 726 (2013).

'Systematic study of (p,γ) reactions on Ni isotopes'. Simon, A., Spyrou, A., Rauscher, T., Fröhlich, C., Quinn, S.J., Battaglia, A., Best, A., Bucher, B., Couder, M., Deyoung, P.A., Fang, X., Gorres, J., Kontos, A., Li, Q., Lin, L.-Y., **Long, A.**, Lyons, S., Roberts, A., Robertson, D., Smith, K., Smith, M.K., Stech, E., Stefanek, B., Tan, W.P., Tang, X.D., Wiescher, M. *Physics Review C* 87 (2013).

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